Group02 - PA2 - User Analysis

Project Information

- **Course:** Human - Computer Interaction
- **Topic:** Running Music Coach - Adaptive Music and Eyes-Free Interaction for Runners
- **Team:** Group02
- **Lecturer:** Le Khanh Duy
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Members

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1. Analysis Goal

This document converts the user research findings into structured design insight through problem identification, brainstorming, affinity analysis, and prioritized problem statements.

2. Input Data

The analysis is based on:

- 7 interview summaries from participants P01-P07;
- contextual observation notes collected during or around running sessions;
- findings, personas, and work models reported in the Group02-PA2-UserResearch document.

3. Problem Identification

From the research data, the team identified recurring user problems:

- music is interrupted by voice coach, notifications, and system ducking;
- users want to monitor pace/effort without repeatedly checking a screen;
- touch interaction is unreliable with sweaty hands and movement;
- smartwatch interaction is better than phone interaction, but still error-prone on small displays;
- some songs unintentionally increase running pace and lead to overexertion;
- users do not fully trust data when synchronization or GPS quality is poor;

- users already create manual workarounds because current tools do not support running flow well.

4. Brainstorming of Potential Solution Ideas

During group brainstorming, the team generated a broad set of ideas before selecting a direction.

4.1 Ideas for Music Flow and Motivation

- Focus Run Mode to suppress non-critical notifications
- Adaptive Music Sync to better match target cadence or effort
- Ambient audio cues instead of spoken coaching
- End-of-segment coaching rather than mid-song interruption
- Lightweight motivation milestones that do not cut music

4.2 Ideas for Eyes-Free Pace Awareness

- Haptic pace feedback with distinct patterns
- Effort-based coaching combining pace, cadence, and heart rate
- One-state visual indicator for fast quick glances
- Beat-to-pace matching through music rhythm
- Just-in-time alerts only after sustained deviation

4.3 Ideas for Reliable In-Run Interaction

- Run Lock interface with large controls
- Earbud gesture integration for quick actions
- Auto-pause / auto-resume smart logic
- One-glance dashboard with only top metrics
- Voice-free quick actions such as double tap or long press

5. Affinity Analysis

To synthesize the research data, the team grouped observations, behaviors, pain points, and idea fragments into thematic clusters using sticky-note affinity mapping on Miro.

5.1 Affinity Diagram

Figure 3. Affinity diagram created from interview and observation notes (Miro)

The Miro board groups research notes into four main clusters plus research context and key selection notes:

- Cluster A: Music & Motivation
- Cluster B: Pace & Cadence (Eyes-Free Monitoring)
- Cluster C: Device Interaction Friction
- Cluster D: Interruptions & Workarounds

5.2 Affinity Summary Table

Cluster	Evidence from users	Related ideas
Music and Motivation	Music keeps solo runners engaged; voice cues break emotion and rhythm	Focus Run Mode, Adaptive Music Sync, Ambient Audio Cues
Pace and Cadence	Users rely on breathing, stride, and body feel; visual checks are disruptive	Haptic Pace Feedback, Beat-to-Pace Matching, Just-in-Time Alerts
Interaction Friction	Unlock, swipe, sweaty fingers, and small watch screens cause errors	Run Lock, Earbud Controls, Voice-Free Quick Actions
Interruptions and Workarounds	Users prepare playlists, mute notifications, disable voice coach, and avoid interaction	End-of-Segment Coaching, Focus rules, Auto Pause logic
Data Trust	Sync mismatches and GPS issues reduce confidence in summaries	Better sync handling, transparent mismatch reporting

5.3 Cluster A - Music and Motivation

Evidence:

- 6 of 7 participants reported that music helps them stay motivated, especially when running alone.
- 4 of 7 explicitly described voice coach or notification audio as disruptive because it ducks music and breaks emotional flow.
- 2 participants said they may feel demotivated or quit earlier without music.

Interpretation:

Music is not only entertainment. It is a core motivational layer that directly affects run continuity.

5.4 Cluster B - Pace and Cadence (Eyes-Free Monitoring)

Evidence:

- 5 of 7 participants rely more on breathing, body feel, stride, or music rhythm than on continuous numeric checks.
- 4 of 7 said looking at phone/watch while running causes distraction, dizziness, or breathing disruption.
- 3 participants described unintentional pace increases caused by fast music tempo.

Interpretation:

Users want feedback, but mostly in an eyes-free or low-attention form.

5.5 Cluster C - Device Interaction Friction

Evidence:

- 5 of 7 participants reported friction from unlock, pause/resume, swipe, or mis-taps, especially with sweaty hands.
- 4 of 7 prefer smartwatch-first interaction to reduce phone handling, but still face small-screen issues.
- 2 participants continue carrying the phone throughout the run because of discomfort or lack of alternatives.

Interpretation:

Current touch interfaces are not designed for high-motion, sweaty, low-attention use.

5.6 Cluster D - Interruptions and Workarounds

Common workarounds:

- preparing playlists before the run;
- enabling Do Not Disturb / Focus mode;
- disabling voice coach;
- leaving phone in pocket and using watch or earbuds instead;
- checking pace only at selected moments;
- accepting reduced interaction to avoid friction.

Interpretation:

The workarounds show strong unmet needs and provide concrete direction for solution design.

6. Selection of Key Clusters

Among the identified clusters, the team selected three as the highest priority:

1. **Music and Motivation**
2. **Pace and Cadence**
3. **Device Interaction Friction**

These clusters were prioritized because they appeared consistently across participants and have the strongest direct impact on running flow.

7. Persona-Oriented Analysis

Persona 1 - Anh (Phone-First Runner)

- depends heavily on phone-based interaction;
- experiences dizziness and disruption from repeated screen checks;
- needs non-visual feedback and simpler in-run controls.

Persona 2 - Binh (Watch-Assisted Runner)

- reduces phone dependency through smartwatch use;
- still suffers from small-screen interaction limits;
- needs subtle feedback rather than a dense metric dashboard.

Persona 3 - Minh (Music-Driven Solo Runner)

- strongly depends on music for motivation;
- is highly sensitive to interruption timing;
- needs just-in-time support that does not break flow.

8. Problem Statements

Problem Statement 1 - Music Flow and Motivation

User: Phone-first runners and music-driven solo runners
Need: A way to preserve motivation and running rhythm without voice coach or notifications cutting into music
Insight: Music is a major motivational resource, and interruption causes emotional and behavioral cost

Statement: Runners who usually run alone with music need a way to preserve motivation and flow because current audio interruptions break rhythm, reduce enjoyment, and can cause early session abandonment.

Problem Statement 2 - Eyes-Free Pace Awareness

User: Phone-first runners and watch-assisted runners
Need: A way to know whether they are running steadily, too fast, or overexerted without constantly looking at a screen
Insight: Visual pace checks are distracting, unsafe, and break breathing rhythm

Statement: Runners need a way to monitor pace, cadence, or effort through eyes-free or low-attention feedback because repeated visual checking interrupts running flow.

Problem Statement 3 - Reliable In-Run Interaction

User: Phone-first runners and watch-assisted runners
Need: A more reliable way to pause, resume, check stats, and control music while moving
Insight: Sweat, vibration, and small touch surfaces make current interaction error-prone

Statement: Runners using a phone or smartwatch need a more reliable in-run interaction method because current touch-based controls are frustrating, error-prone, and disruptive to the workout.

9. Prioritized Insights

1. Protect music continuity first because motivation and mood depend on it.
2. Provide eyes-free pace cues because constant visual monitoring is costly.
3. Reduce touch complexity because current interaction reliability is low in running conditions.
4. Design around real user workarounds instead of forcing more alerts and more voice.

10. Design Requirements Derived from Analysis

- Feedback should prioritize haptic or ambient channels over spoken interruption.
- Alerts should be just-in-time rather than constant.
- Core in-run actions should require minimal steps and large touch targets.
- Non-critical notifications should not hard-cut music during active running.
- The solution should support both phone-only and watch-assisted runners.
- Sync and data trust issues should be handled transparently where possible.

11. Analysis Conclusion

The analysis shows that the strongest design opportunity is to support running flow by combining adaptive music behavior, subtle pace feedback, and reduced mid-run interaction burden. This conclusion directly informs the proposal and use case specification documents.